

Element D

Morphological Chart:

Functions	Design #1	Design #2	Design #3	Design #4
Change color	Utilize a remote	Use buttons attached to the device	Utilize a smartphone app	Utilize Remote
Fit into the sensory room	Attach to a wall	Tuck into one of the existing cubbies	Make the device a dark color to fit brightness of the room	Tuck into an existing cubby
Control the brightness	Utilize Remote	Utilize a smartphone app	Use buttons	Utilize Remote
Have flashing lights/pattern	Color-changing gradient	Programmed color changing	Brightness changing pattern	Programed color changes
Create light	Use LED lights	Use LEDs and fiber optic cables	Use fiber optic cables	LED Strips
Have optional interactivity	Visual stimulus from patterns and color changing	Interact with buttons that correspond with lights	Have the ability to touch fiber optic cables	No interactivity

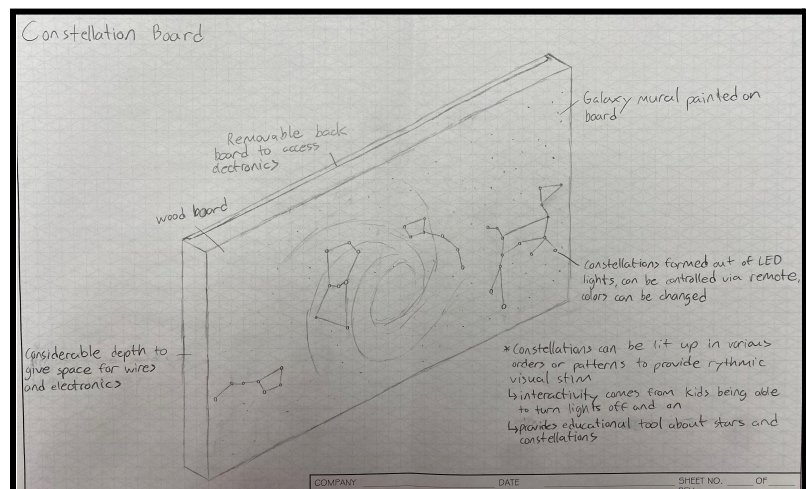
Conceptual Designs:

Constellation Board:

Advantages

- Interactive
- Multiple functions (Can be used as a reward, game, educational tool, calming device)
- Visually stimulating (lights, cause and effect)

Trade-Offs

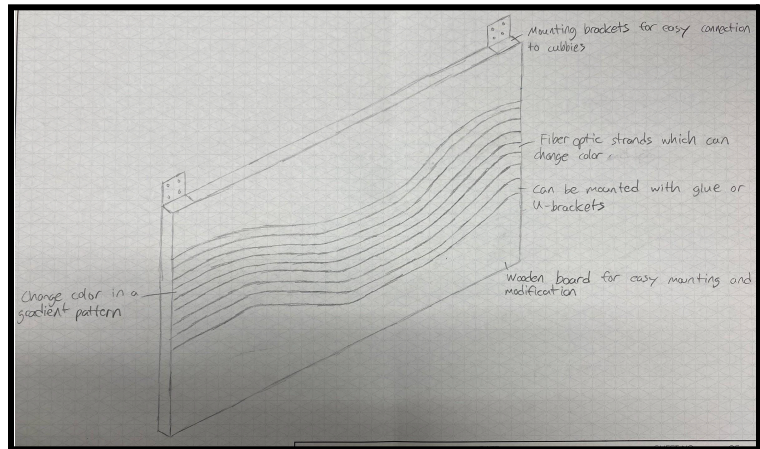


- Very complex design and function
- Expenses
- Could not fit the classroom “niche”

Fiber Optic Wave Board:

Advantages

- Passive calming effect through the color changing gradient, same effect as the bubble tube
- Simple to design and construct
- Colors can also be changed manually for “mood lighting”
- Can be any size depending on space availability (will have to ask Ms. Brennan)
- Not too many electronics involved so easy maintenance and implementation into the room



Trade-Offs

- No interactivity whatsoever (may be a good thing depending on the kid)
- Fiber optics are extremely expensive
- Will be difficult to figure out how to work and program them
- Lack of research on fiber optics currently

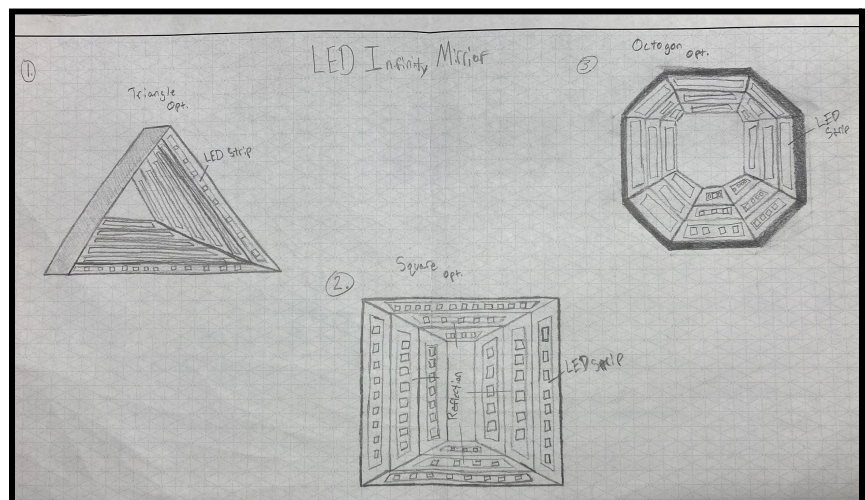
Led Infinity Mirror

Advantages

- Adjustable lighting can create different moods
- Mesmerizing and calming effect
- Simple Construction
- Passive stimulation can appeal to some kids

Trade-Offs

- “Infinity” effect can be



unsettling to some children

- Lack of interactivity
- Glass could break and become a hazard
- Glass could easily be scratched

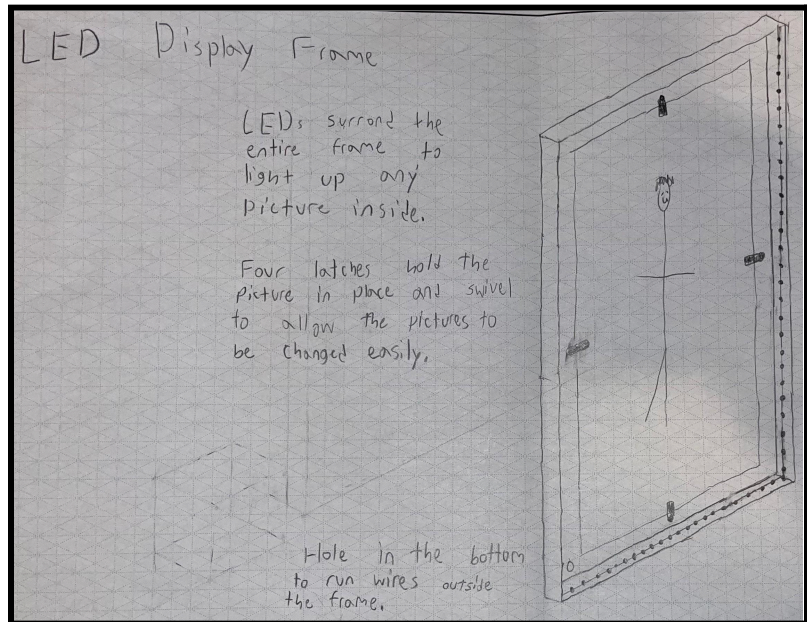
LED Display Frame:

Advantages

- Artwork can easily be changed
- Reward system, they draw something, they get to put it in the board
- Easy to build
- Lights can be changed easily with a remote
- Good way for kids to have direct involvement through being able to put their pictures in display

Trade-offs

- Not all of the kids are interested in art
- Kind of a boring device that may not provide any relaxation



Decision Matrix:

		Constellation Board		Fiber Optic Wave Board		LED Infinity Mirror		LED Display Board	
Priority #	Weight	Score	Weighted Score	Score	Weighted Score	Score	Weighted Score	Score	Weighted Score
1	30%	7	2.10	9	2.70	7	2.10	7	2.10
2	25%	5	1.25	7	1.75	8	2.00	8	2.00
3	20%	7	1.40	6	1.20	7	1.40	6	1.20
4	15%	8	1.20	6	0.90	9	1.35	8	1.20
5	10%	9	0.90	10	1.00	3	0.30	3	0.30
	100%		6.85		7.55		7.15		6.80

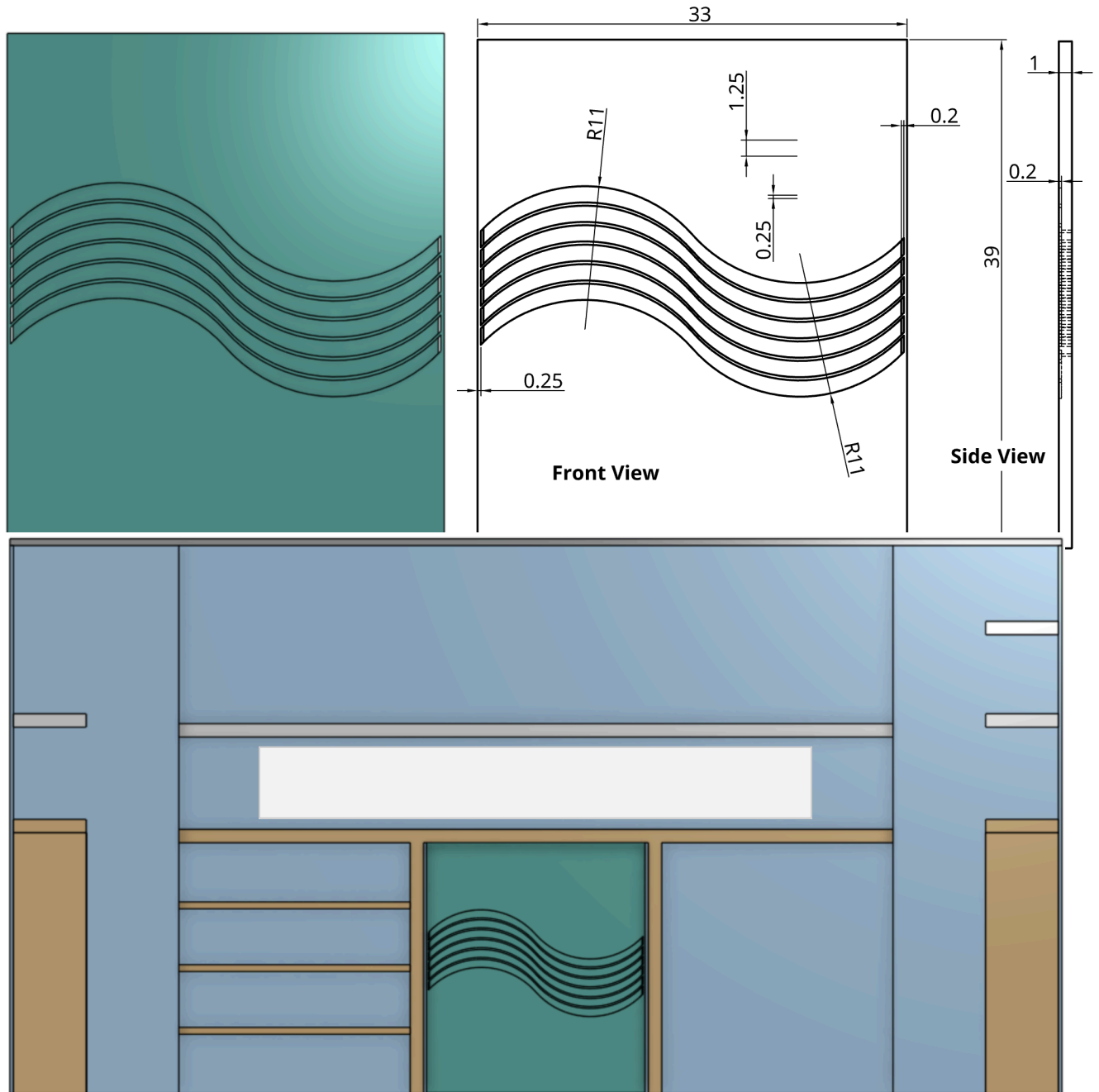
Design Solution Choice:

We chose to go with the Fiber Optic Wave Board because not only did it achieve the highest score on the decision matrix with a score of 7.55, but it also effectively combines all of the positive elements of the other designs. It's extremely easy to construct as it only has two main parts, and it's also easy to use because the fiber optic lights are pre-programmed to fade in and out. Furthermore, the lights can easily be controlled using a remote that they come with, or using an app that can be downloaded on the teachers' phones. Unlike the other designs, the wave board doesn't utilize any glass or small parts, so it's extremely safe and doesn't run the risk of easily breaking and injuring the kids. Also, Mrs. Brennen told us that it was her favorite. Another reason we chose fiber optics is because of the amount of variation in the color of the lights and the speed of the flashing. This gives the kids plenty of options to keep them stimulated, and provides new experiences to keep them from getting bored of the device. Being able to control the speed of the flashing allows it to be personalized to each student. Additionally, Ms. Brennan specified that she prefers it not having any interactivity so that it can effectively provide strictly visual stimulation, which has benefits not tied to touch interactivity.

Plan of Action:

Phase	Action Completed
1	Gather all necessary materials (fiber optic strands, wood, hardware, etc.)
2	Create and finalize an onshape drawing of the wooden backboard of the project
3	CNC the wooden board
4	Install power source on the back of the board, and install 8 fiber optic strands/groove onto the board utilizing hot glue.
5	Install board into the cubby at WES utilizing L brackets that connect the board to the wooden cubby walls
6	Plug the power cord into the outlet below the cubby and make sure the cord is out of reach of the kids

Design Blueprint:



Project Budget

Item	Price
Fiber Optic Device	\$105.59
Plywood - 4' x 8'	\$50
L Brackets	\$8.48
Total	\$164.07